

Question	Answer	Marks
6(a)	mass of α (particle) is <u>much</u> greater (than β^+ particle)	B1
	both particles are positively charged	B1
	(magnitude of) charge on α (particle) is twice the charge (on β^+ particle)	B1
6(b)(i)	cross labelled Q at (82, 212)	B1
6(b)(ii)	particles emitted are: • beta-minus (particle) / electron • (electron) antineutrino either particle named	B1
	both particles named and no incorrect particles named	B1
6(c)(i)	$212(u) \times 1.3 (\times 10^5) \sin 68^\circ$ or $4(u) \times 150 (\times 10^5) \sin \theta$	C1
	$4(u) \times 150 (\times 10^5) \times \sin \theta = 212(u) \times 1.3 (\times 10^5) \times \sin 68^\circ$	C1
	$\sin \theta = 0.426$ $\theta = 25^\circ$	A1
6(c)(ii)	$E = \frac{1}{2}mv^2$	C1
	$= \frac{1}{2} \times 4 \text{ u} \times (150 \times 10^5)^2$ $= \frac{1}{2} \times 4 \times 1.66 \times 10^{-27} \times (150 \times 10^5)^2$ $= 7.5 \times 10^{-13} \text{ J}$	A1